

Theoretical Assessment Questions for the 2017 International

Biology Olympiad

National Team Selection **Camp**

Part One

Student Code



Precautions:

1. This section of the test consists of 9 pages (including the cover) and is worth 100 points. Please check if you have it.

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4. You may write your answers on the back of the questions, but please indicate the question number.

Part 1: (100 points)

I. International Competition Type Questions (Please answer this section on the answer sheet) Questions

1-13 below are international competition type questions, with unequal scoring (4 points per question, 52 points in total; 0 points for 0 correct answers, 0.2 points for 1 correct answer, 1 point for 2 correct answers, 2 points for 3 correct answers, and 4 points for 4 correct answers).

1. Toll protein was first discovered in *Drosophila* embryos. Its function is to regulate the development of the dorsoventral body axis. The lack of toll protein causes abnormal development in *Drosophila* embryos. In 1996, Jules Hoffmann discovered a *Drosophila* with a mutated toll gene that, due to a lack of signal translation, could not synthesize drosomycin and was thus susceptible to fungal infection. It was later confirmed that toll protein can recognize the specific structure of pathogens, triggering the *Drosophila* to secrete polypeptides to resist infection. In 1997, Charles Janeway et al. discovered a human toll protein similar to the *Drosophila* toll protein in human immune cells. Subsequently, many similar molecules were discovered, all possessing similar structures and receptor functions, and were named toll-like receptors (TLRs). TLRs are an ancient protein family; tobacco TLRs help resist tobacco mosaic viruses, making them arguably the front line of biological resistance against pathogen invasion. Is this description correct (T) or incorrect (F)?

- (A) TLR can initiate an innate immune response, but is not related to the initiation of an adaptive immune response.
- (B) TLR utilizes signaling pathways common in both the plant and animal kingdoms to initiate infection control.
- (C) TLR can promote the evolution of multicellular organisms; those unable to resist infection cannot survive. TLR plays a role in the differentiation of cells. Similarity in substructures provides a way to trace the origins of life.
- (D) There are 10 known TLRs in humans. Each TLR can recognize a specific basic component of a certain type of pathogen. Because the basic components of the same type of pathogen are the same, just 10 TLRs are enough to defend against almost all known pathogens. These TLRs can also work in groups, and different types of combinations can exist on different cells.

2. Natural killer (NK) cells originate from lymphocyte precursor cells and account for 5-10% of lymphocytes in human blood. NK cells lack specific antigen receptors on their surface and are primarily responsible for non-specific defense. Their cell surface has inhibitory and activating receptors. When in contact with normal cells, the type I MHC molecules on the normal cell surface bind to the inhibitory receptors, transmitting a message that inhibits NK cell activity, and the cell is not activated. However, after cells are infected by viruses or transform into tumor cells, the expression of type I MHC molecules often becomes abnormal. When these cells come into contact with NK cells, they cannot receive the message from the inhibitory receptors, and thus the NK cells are activated and perform cell killing. NK cells will only perform cell killing when the activation message is present and the inhibitory message is abnormal or absent. Please answer whether the description is correct (T) or incorrect (F)?

- (A) NK cells are derived from myeloid precursor cells.
- (B) NK cells have receptors for IgG molecules on their surface, which can kill target cells with the assistance of IgG.
- (C) NK cells have many membrane proteins on their surface that can receive stimuli from target cells and regulate NK cell activity.
- (D) NK cells can use perforin to create pores in the target cell membrane, allowing granzymes to enter. (granzyme) enters target cells and alters the permeability of their mitochondrial membranes, initiating apoptosis to kill tumor cells and virus-infected cells.

3. When the immune system is abnormal or damaged, and the organism is unable to fight off pathogens, it is called immunodeficiency. Immunodeficiency is divided into congenital and acquired types. Congenital immunodeficiency is caused by defects in genes or the development of the immune system, and is generally classified into four categories: phagocytic immunodeficiency, complement deficiency, antibody deficiency, and cellular immunodeficiency. Acquired immunodeficiency is caused by acquired factors such as viral infection, chemotherapy, and immunosuppressants, leading to immune deficiency. Currently, the most serious form is AIDS. Please answer whether the description is correct (T) or incorrect (F)?

- (A) One of the characteristics of phagocytic leukocyte deficiency is an increased risk of bacterial infections.
- (B) Congenital agranulocytosis is caused by a defect in lymphoid precursor cells.
- (C) Patients with DiGeorge syndrome have congenital thymic hypoplasia and a lack of T cells in their blood.
- (D) Both HIV-1 and HIV-2 require a common CCR5 receptor to enter T cells; therefore, HIV-1 and HIV-2...
The correlation is higher than that between HIV-1 and SIV.

4. Cells in an organism generally die through either apoptosis or necrosis. Is this description correct (T) or incorrect (F)?

- (A) Cell culture may induce apoptosis if oxygen or nutrients are lacking, or if the pH or temperature is altered.
- (B) During apoptosis, the cell structure dissolves, the cell membrane is destroyed, organelles swell, and chromatin disperses into flocculent particles.
- (C) Apoptosis involves autophagy, gene regulation, synthesis of new proteins, appearance of apoptotic bodies, and cell turnover.
The increase in intercellular Ca^{2+} is observed during necrosis, but these phenomena do not occur during necrosis.
- (D) CD4 and CD8 double-positive T cells in the thymus undergo two types of colony selection: (1) apoptosis of double-positive T cells with weak MHC molecule recognition ability because they cannot recognize antigens, and (2) apoptosis of double-positive T cells that will produce an immune response to self-antigens to prevent them from causing autoimmune diseases.

5. Dendritic cells (DCs) are few in number in the blood, but are considered the most antigen-presenting cells in the immune system and key cells linking innate and adaptive immunity. Is this description correct (T) or incorrect (F)?

- (A) When immature dendritic cells mature, their phagocytic capacity increases.
- (B) Dendritic cells can originate from bone marrow stem cells or lymphoid stem cells, and the distribution of dendritic cells from different origins is...
Location and function will also differ.
- (C) Some dendritic cells, after forming in the bone marrow, can enter the epidermal layer of the skin via the bloodstream and are called Langerhans cells.
- (D) Dendritic cells can acquire pathogen components during the innate immune response, transporting them from the site of infection to the lymph nodes. They then present antigen fragments to T cells via type II MHC molecules on their cell membranes, leading to...
T cell activation leads to an adaptive immune response.

6. *Naegleria fowleri** (*Nf*), commonly known as the brain-eating amoeba, has stages in its life cycle including trophozoite, cyst, and flagellate. The trophozoite can transform into a flagellate under unfavorable conditions; it swims rapidly and can revert to a trophozoite within 24 hours, but cannot form a cyst. It reproduces most vigorously at 42°C.

Nf can travel along the olfactory nerve into the brain and cause primary amoebic meningoencephalitis (PAM), which has a mortality rate of up to 99%. Please answer whether this description is true (T) or false (F)?

- (A) *Nf* is a parasitic species that lives in the human brain.
- (B) *Nf* infection in humans also occurs through ingestion of food (or water) contaminated with trophozoites.
- (C) If amoebas can cross the blood-brain barrier and enter the human brain, they will trigger PAM.
- (D) The trophozoite is thermophilic and represents the pathogenic stage in the life cycle; it can form cysts when encountering adverse conditions.

7. Immune responses involve complex interactions between hematopoietic cells, inflammatory cells, lymphocytes, and other cells.

Cytokines play a crucial role in intercellular communication. Cytokines are low-molecular-weight regulatory proteins secreted by leukocytes and other cells in response to various stimuli. These proteins assist in regulating the development of functional immune cells and other direct effector effects. Is this description correct (T) or incorrect (F)?

- (A) Lymphokines are secreted only by lymphocytes, while monokines are cytokines secreted by monocytes and macrophages.
- (B) Cytokines can bind to specific receptors on the cell membrane of target cells, triggering signal transduction pathways and thereby altering gene expression in the target cells.
- (C) When the combined effects of two cytokines on cellular activity are better than the additive effects of individual cytokines, this is called cytokine antagonism.
- (D) Cytokines secreted by activated T cells can affect the activity of B cells, Tc cells, natural killer cells, macrophages, granulocytes, and even hematopoietic stem cells.

8. The following statements about the cell nucleus are correct (T) or incorrect (F).

- (A) The nuclear membrane consists of two membrane-like structures that overlap in a parallel manner, with a gap between the inner and outer membranes, and a smooth inner surface.

The lumens in the plasma reticulum are connected

- (B) Nuclear pores that penetrate the nuclear membrane have a channel-like function.
- (C) Biomolecules pass through nuclear pores via karyopherin, a molecular mediator.
- (D) Importins and exportins help transport biomolecules from the cell nucleus.

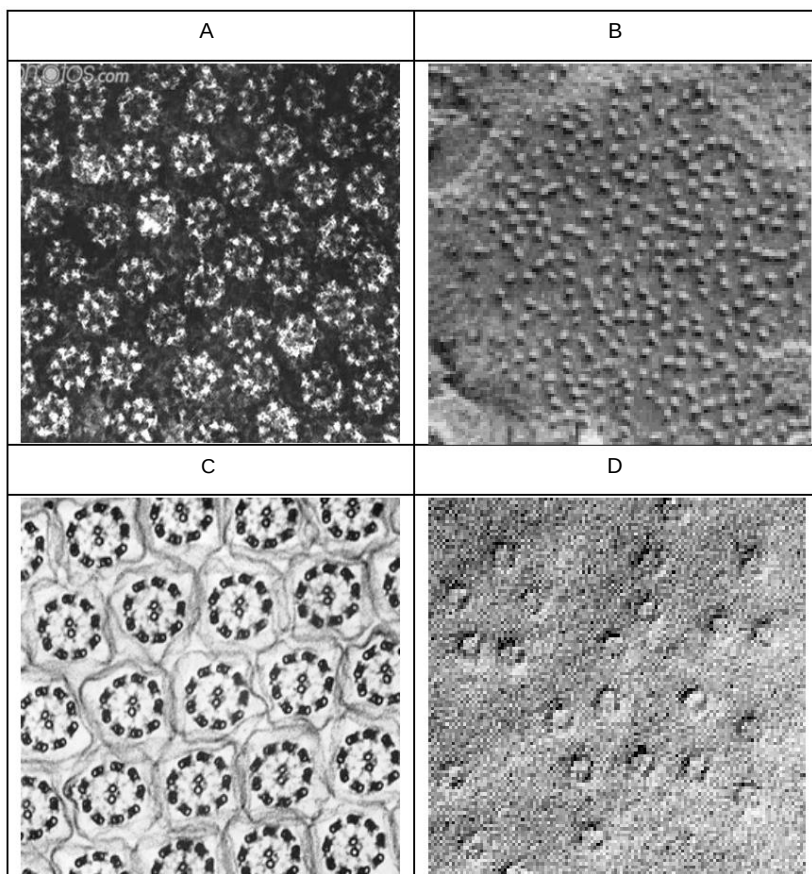
Gogli apparatus

9. Which of the following biomolecules can enter the cell nucleus through nuclear pores to perform biological functions?

Is the description correct (T) or incorrect (F)?

- (A) Transcription factor
- (B) Messenger RNA (mRNA)
- (C) Glucose molecule
- (D) Albumin molecule

10. Which of the following is an electron microscope image of a nuclear pore? Please answer whether the description is correct (T) or incorrect (F)?



11. The Golgi apparatus was coined in 1898 by the Italian anatomist Camillo Golgi.

Golgi discovered and named it. Regarding the functions of Golgi, please answer whether the following descriptions are correct (T) or...

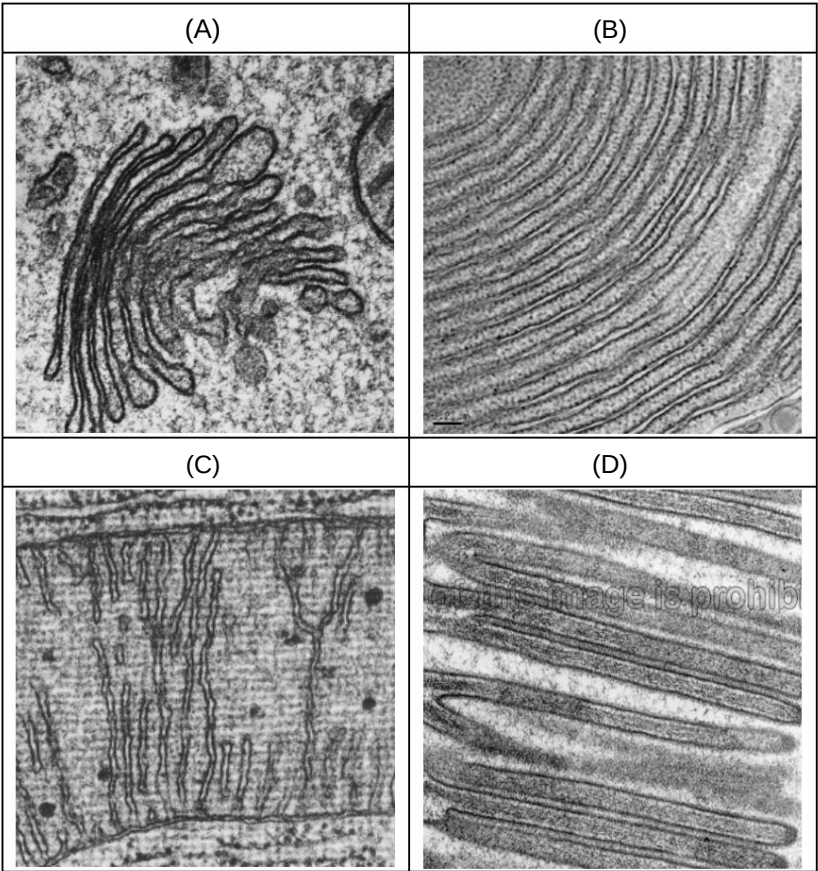
Error (F)?

- (A) Related to protein processing and sorting
- (B) Related to protein breakdown
- (C) Related to protein secretion
- (D) Related to cellular respiration

12. In which of the following cell types are Gauden bodies most developed? Choose the correct answer (T) for those with developed Gauden bodies and the incorrect answer (F) for those with underdeveloped Gauden bodies.

- (A) Plasma cell
- (B) Thyroid cells
- (C) Nerve cells
- (D) Cardiomyocytes

13. Which of the following is an electron microscope image of a high-strength structure? Please answer whether the description is correct (T) or incorrect (F)?



II. Short Answer Questions (29 **points in total**)

1. Compare the immune responses of arthropods and humans **(4 points)**

2. Taiwan has recently experienced an outbreak of avian influenza, resulting in the mass deaths of many poultry; this wave of avian influenza is caused by type A H5N6.

Caused by highly pathogenic avian influenza virus subtypes, human infection has an extremely high mortality rate, impacting health and safety as well as social and economic factors.

The relief is enormous. May I ask...

- (1) What is a highly pathogenic avian influenza virus? **(5 points)**

- (2) What do the H and N in the H5N6 virus code name refer to respectively? **(5 points)**

3. Archaea and eubacteria both belong to the prokaryotic cell family in terms of cell structure.

If we disregard ecological differences, please describe the differences in cellular structure between the two organisms. (8 points)

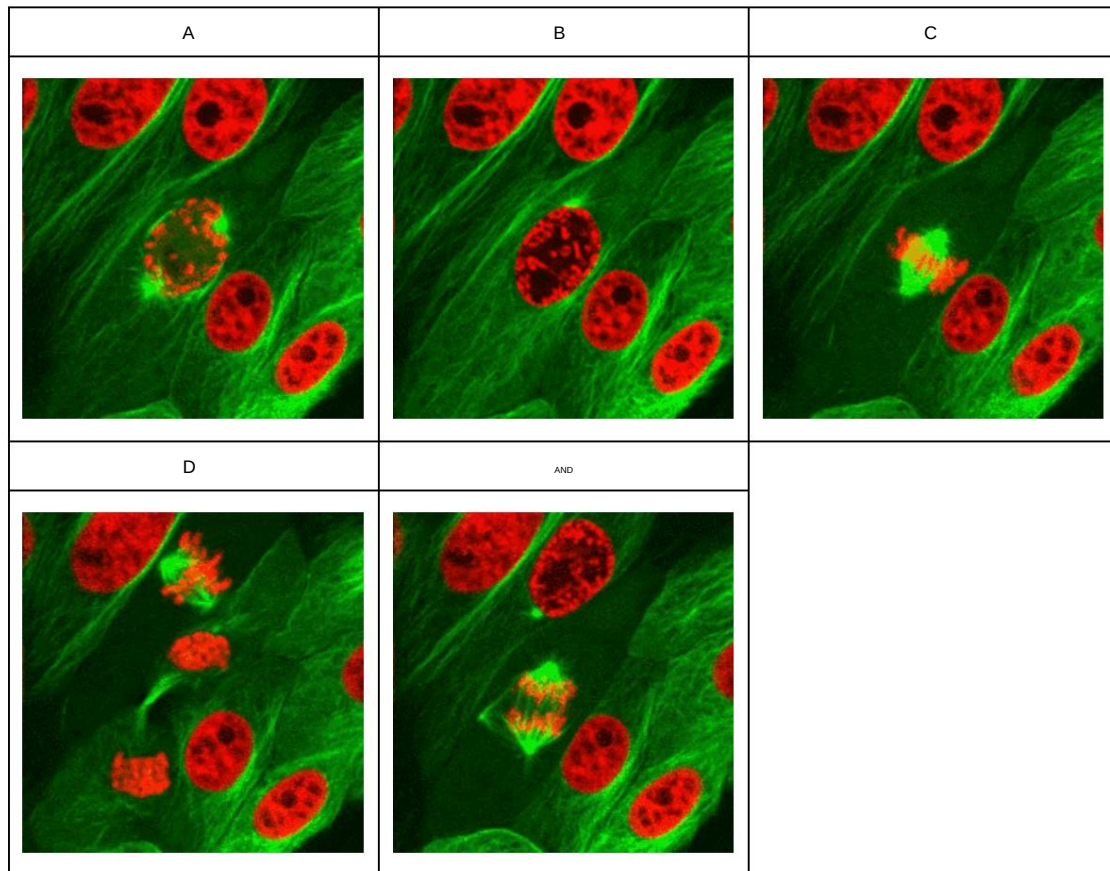
4. Please compare the differences in structure and movement between bacterial flagella and eukaryotic flagella. (7 points)

III. Fill in the blanks (19 points in total)

1. Xiaoming used time-lapse photography to observe and record the process of cell division.

Do not use green or red fluorescent dyes for the labeling and staining of related biomolecules. The results are shown in Table AE below.

Show.



1.1 What biomolecules do red and green fluorescent dyes stain respectively?

(No points will be awarded for any incorrect answer) (2 points)

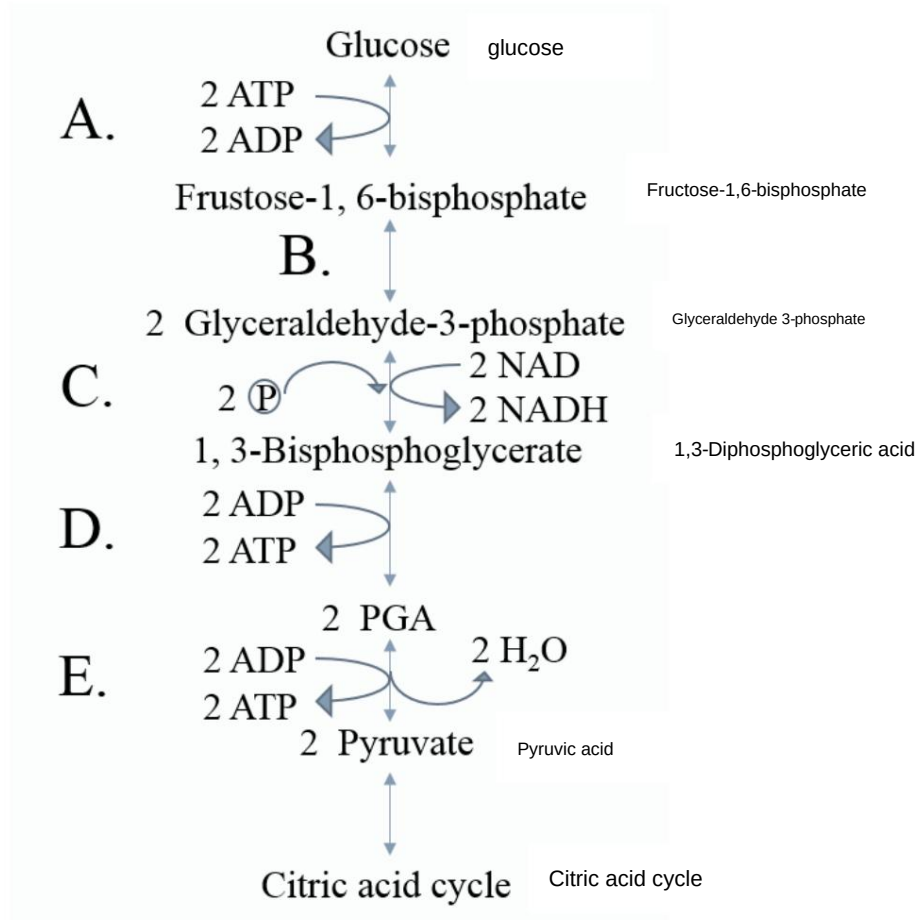
green	red

1.2 Please fill in the results of AE in the table below in the order of cell mitosis.

(No points will be awarded for any incorrect answer) (5 points)

Early stage	Early to mid-term	Mid-term	Later	Late period
prophase	prometaphase	metaphase	anaphase	telophase

2.



The diagram above illustrates the steps of glycolysis. Each step is represented by a letter. Please use these letters to answer the following questions.

The problem.

2-1. What steps show that a molecule breaks down into two smaller molecules? (2 points) _

2-2. Which step shows that the inorganic phosphate has been added to the reactants? (2 points) _

2-3. What steps demonstrate the phosphorylation reaction, and that the phosphate source is ATP? (2 points) _

3. The following are the basic steps of meiosis. Please answer the following questions based on these steps.

- a. Four new nuclei are formed, each containing half the chromosomes of the parent.
- b. The tetrad is located on the equatorial plate.
- c. Separation of sister chromosomes
- d. Separation of homologous chromosomes
- e. Synapsis; the movement of chromosomes in pairs into the middle of the cell.

3-1. Based on the above description, what is the order of meiosis? (3 points) _

3-2. Which of the above steps is also observed in mitosis? (3 points) _

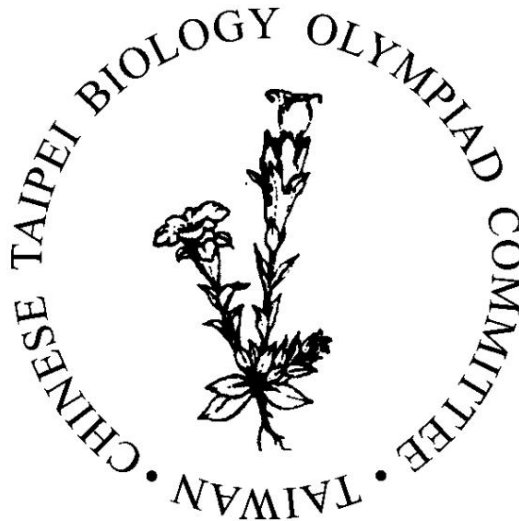
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Part Two

Student Code



Precautions:

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Part Two: (Total 100 points)

1. The following are relevant statements about plant seed germination:

Please fill in the correct corresponding description codes (A~Z) in the right column for items 1~8 in the left column of the table below.

Fill in the blanks with the corresponding descriptive codes. (2 points each, **16 points in total**)

Seed germination	(Enter the description code)	Description code			
1. Environmental factors that affect seed germination		A water	L cytokinin		
2. Hormones that affect seed germination		B Oxygen	M Seed coat too thick		
3. Germination of corn seeds on the ear is due to...		C Temperature	N Seed coat contains inhibitory factors		
4. Possible reasons why seeds fail to germinate		D Wind force	O Absorption acid too low		
5. Photoreceptors that affect seed germination		E Light or Darkness	P High Jasmonic Acid		
6. When conditions are suitable for seed germination		F Auxin	Q Cryptochrome		
7. Enzymes involved in seed germination		G Gefibrin	R Photosensitizer	A	
8. Provides a source of nutrients for seed germination.		H Jasmonic acid	S Photosensitive	B	
		I. Salicylic acid	T decreases	turgor pressure	
		J Ethylene	U swell	pressure increases	
		K-abscisic acid	V oxidase		
			W γ-hydrolase		
			X	γ-hydrolase	
			Y	Fat molecules	
			Z	starch	

2. The following information pertains to the effects of light and darkness treatments on flowering and gene expression, as well as plastid (or chloroplast) development.

Based on the relevant descriptions of education, please answer the following questions according to the table and diagram below. (Total 34 points)

(Table 1): The flowering time of wild-type Arabidopsis (Columbia) and four other different mutants (*co-1*, *gi-2*, *ld-1*, and *hy2*) was investigated under continuous light and continuous darkness. The flowering time was recorded as days to anthesis.

The number of leaves when the flower stalk emerges is used to indicate this.

Strain	In continuous light		In continuous darkness	
	Days to anthesis	Leaves	Days to anthesis	Leaves
實驗一				
Columbia	20.4 ± 3.6	7.8 ± 0.84	30	6.4 ± 0.54
<i>co-1</i>	26.4 ± 4.6	11.8 ± 1.6	30	6.2 ± 0.44
<i>gi-2</i>	39.6 ± 6.7	24.2 ± 5.4	30	6.8 ± 0.44
<i>ld-1</i>	60.4 ± 9.3	45.4 ± 6.0	41	10.4 ± 2.1
實驗二				
Columbia	20.4 ± 5.5	7.6 ± 0.55	30	7.6 ± 0.55
<i>hy2</i>	16.4 ± 4.0	4.0 ± 0.70	30	7.6 ± 1.14

(Table 2): Results of gene expression quantification based on Figure 1. The leftmost column shows the genes (*rbcS*, ...) induced by different light exposures.

The gene expression levels of *cab*, *chs*, *rbcL*, *psa A* in samples grown in light (Light-solid(1)) were considered as...

100, other samples: Dark-liquid 3 with shaking, and dark solid culture.

(Dark-solid 2) represents the gene expression level relative to those in the light.

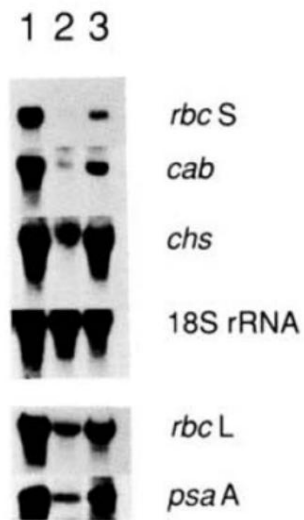
Genes	Light-solid (1)	Dark-liquid (3)	Dark-solid (2)
<i>rbc S</i>	100	20	>1
<i>cab</i>	100	18	1.5
<i>chs</i>	100	70	15
<i>rbc L</i>	100	24	11
<i>psa A</i>	100	36	7

(Figure 1): The expression level of light-induced gene mRNA in different samples.

Lane 1: Arabian mustard growing in light.

Lane 2: Arabian mustard grown on solid culture medium in the dark.

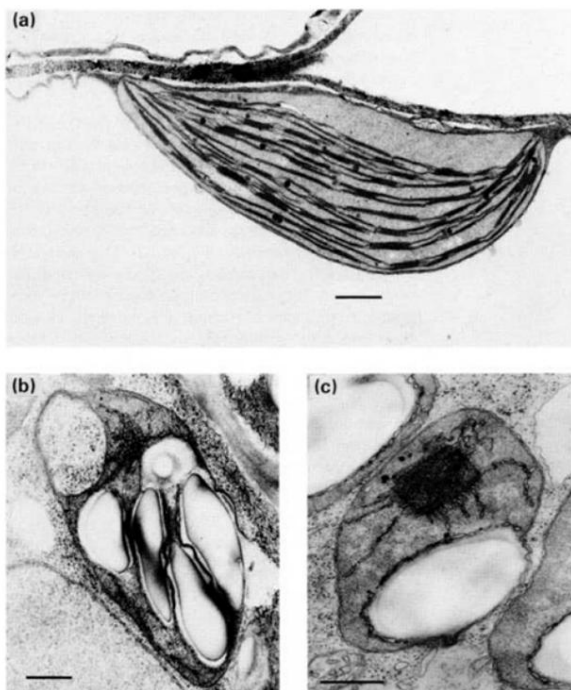
Lane 3: Liquid culture medium in the dark containing agitated Arabica mustard.



(Figure 2): Electron microscopy of chloroplasts or plastids of Arabian mustard plants grown in liquid culture medium in light or darkness.

Mirror image. 15-day-old plants were used for observation. (a) Plastids of a plant growing in light. (b and c) Dark liquid.

plastids of plants grown in a culture medium.



(1) In Table 1, what are the differences in flowering time between the four mutants in light and darkness compared to the wild type?

(10 points)

(2) In Table 1, compare the mechanisms by which plants grow in light and in darkness regulate flowering. Are they the same?

(2 points) Why? **(3 points)**

(3) Based on Figure 1 and Table 2, what conclusions can be drawn by comparing the performance of light-regulated genes ? **(5 points)**

Explain the possible reasons. **(2 points)**

(4) Based on Figure 2, compare the differences in plastid development in figures (a), (b), and (c) **(3 points)**. (b) and (c)

What is the clearly visible, huge, white, oval structure? **(3 points)**

(5) Based on the results in Tables 1 and 2, and Figures 1 and 2, propose the effects of light and darkness on flowering.

The conclusion is drawn, and possible reasons are inferred.

in conclusion:

(4 points)

reason:

(2 points)

3. Draw a phylogenetic tree to illustrate our current understanding of bryophytes.

(Bryophyte), Lycophyte, Pteridophyte, Gymnosperm

The evolutionary relationship between Gymnophytes and Angiosperms. Please use Charophytes as an example.

As an outgroup, it identifies at least one derived characteristic in the system.

On each clade, that is, the feature described thereafter occurs after the common ancestor of the clade node.

Each organism is unique. (10 **points**)

4. Parasitic plants (P) are plants that must obtain some or all of their nutrients from a host plant (H).

Of all the plant species that require all nutrients, approximately 4,100 are known parasitic plants that parasitize 1,900 flowering host plants.

Parasitic plants, in general, possess haustoria, which they use to extract water and nutrients from their host. (Total)

(30 points)

(1) Parasitic plants, based on their relationship with their host, can be categorized according to whether or not they must depend on a host to complete their life cycle.

Parasites are classified as either absolute parasites or facultative parasites based on their life cycle.

Parasites are classified as stem parasites or root parasites based on whether they grow on the main parts of the host; and as hemiparasites based on whether they can perform photosynthesis under natural conditions.

Or holoparasite.

i) Fill in the blanks with either "O" (correct) or "X" (incorrect). (0.5 points per blank, 3 **points total**)

Holoparasites must be absolute parasites.		Absolute parasites must be holoparasites.	
Facultative parasites must be hemiparasites.		Hemiparasites must be facultative parasites.	
Stem parasites must be hemiparasites.		Root parasites must be holoparasites.	

ii) Please describe the characteristics of the following six parasitic plants and create a table comparing their similarities and differences. (9 points)

Cuscuta (stem holoparasite)

African egret flower (*Hydnora* spp.): root holoparasites

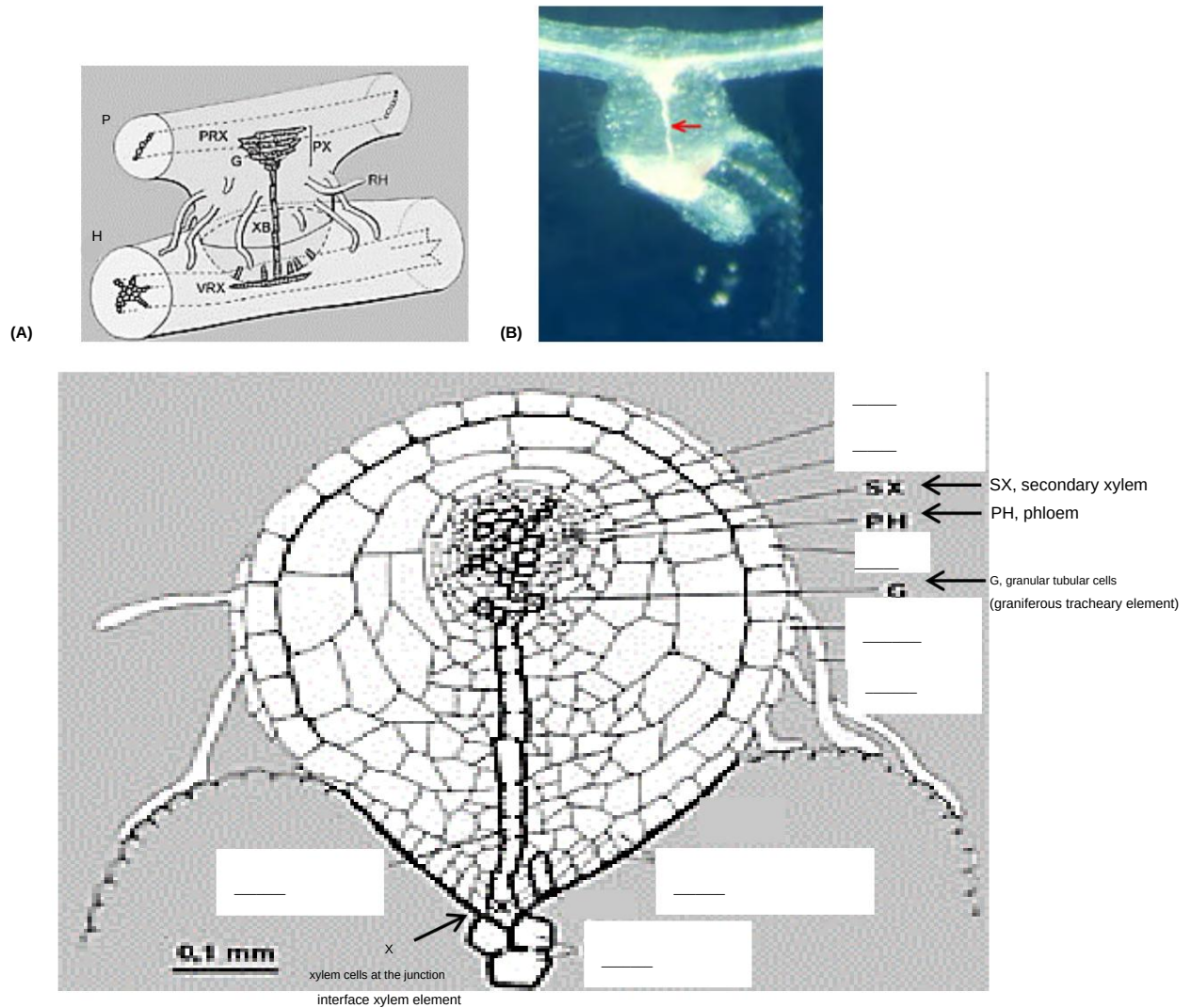
Mistletoe obligate stem hemiparasite

Australian Christmas Tree (*Nuytsia floribunda*)): obligate root hemiparasite

Orobanche: root holoparasites

ýý(*Rhinanthus*) ýfacultative root hemiparasite

- (2) The following are sections of the root structure between a certain Scrophulariaceae parasitic plant and its host plant. (A) is a 3D stereoscopic image; (B) is an autofluorescence image of the section; (C) is a cross section of the haustorium of the parasitic plant and a histological diagram between it and its host.



- i) In Figure A, how many stellar rays (arches) are shown in the cross-sections of the roots of the parasitic plant (P; above) and the host plant (H; below)? What type of tissue are the cells at the tips of the stellar rays? What types of cells might be directly connected between the haustorium of the parasitic plant and the host plant? (6 points)

- ii) What kind of tissue is the spontaneously fluorescent image indicated by the red arrow in Figure B? What is the main substance that produces its fluorescence? (4 points)

iii) Please select the correct names from the following cell names and fill them into the blanks in Figure **C** in order. (8 **points**)

E, epidermis	T, root hair
En, endodermis; HX, host xylem element.	
Pc, pericycle	IP, interface parenchyma cell.
S, exodermis	XB, Vessel member of single-stranded xylem bridge .

5. Which tissues connect the shoot system and root system of a plant?

Why do the stem and root systems of different plants differ in size and morphology, and why do they vary?

This also explains how continuous, indeterminate growth in plants enhances phenotypic plasticity.

(phenotypic plasticity)(10 **ŷ**)

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Part Three

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Part Five: (100 points)

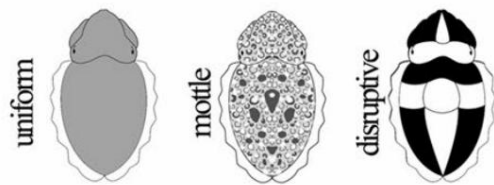
1. True or False Questions (2 points each , 10 points in total)

Cuttlefish camouflage their color and shape in different substrate environments, and researchers can roughly categorize their color camouflage into...

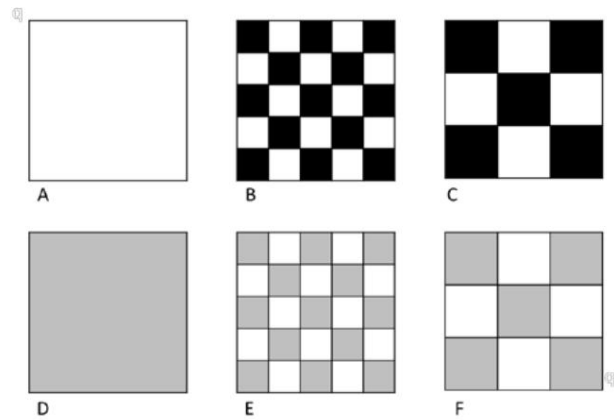
Categories: uniform, mottled, and broken (as shown in Figure 1). Twenty squid of similar size and color from the same batch...

The samples were placed in aquariums with six different patch background colors (Figure 2, A to F), and the following data were obtained:

	A	B	C	D	AND	F
Uniform type 20, Mottled		0	0	19	0	0
type 0, Broken type 0		18	1	1	10	5
		2	19	0	10	15



(Figure 1): The camouflage colors of three types of squid



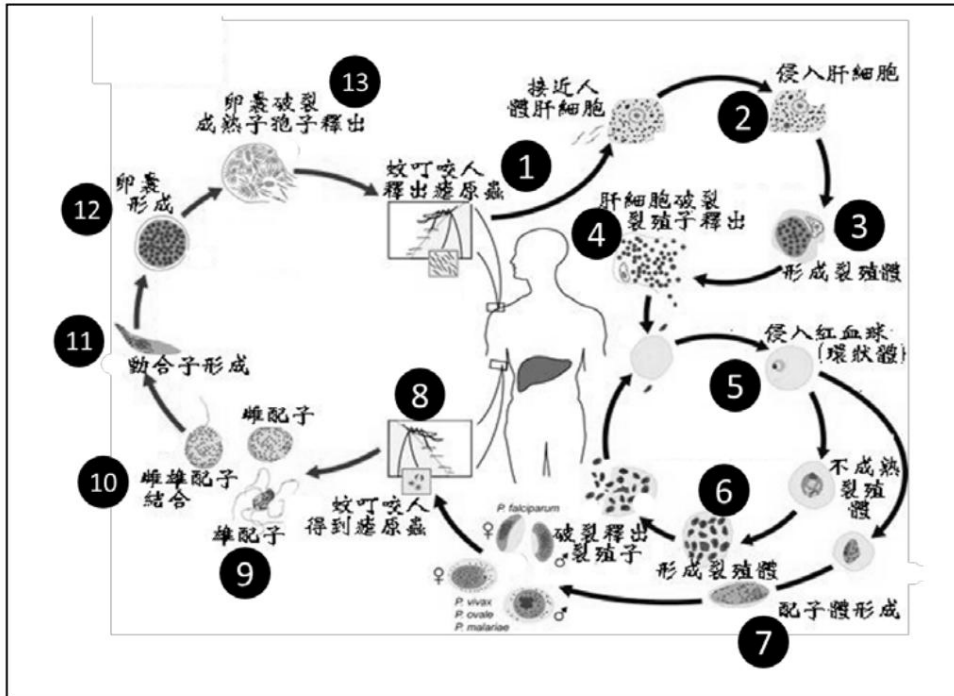
(Figure 2): Six different patch background colors in the aquarium (A to F)

Based on the experimental data above, answer whether the following statements are correct:

- A. (B. The frequency of uniform patches is independent of patch color contrast. Smaller
- (C. (D. patches have a lower frequency of fragmented patches. Greater patch
- (E. (color contrast leads to a higher frequency of mottled patches. Compared to patch size,
- patch color contrast has a greater impact on the frequency of mottled patches.
- Compared to patch color contrast, patch size has a greater impact on the frequency of fragmentation.

II. Fill in the blanks (A-E, 3 points each, 15 points in total)

Please refer to the following simplified diagram of the life cycle of Plasmodium and answer the following questions:



Based on the different morphologies, infection sites, and hosts in the life cycle of Plasmodium, it can be divided into 1 to 13 groups.

During this period, please tell me:

- Which stages of the malaria parasite are haploid? (_____)
- During which periods does the asexual reproduction of Plasmodium occur? (_____)
- Typical symptoms of human malaria include periodic fever, chills, and cold sweats. What are the symptoms?

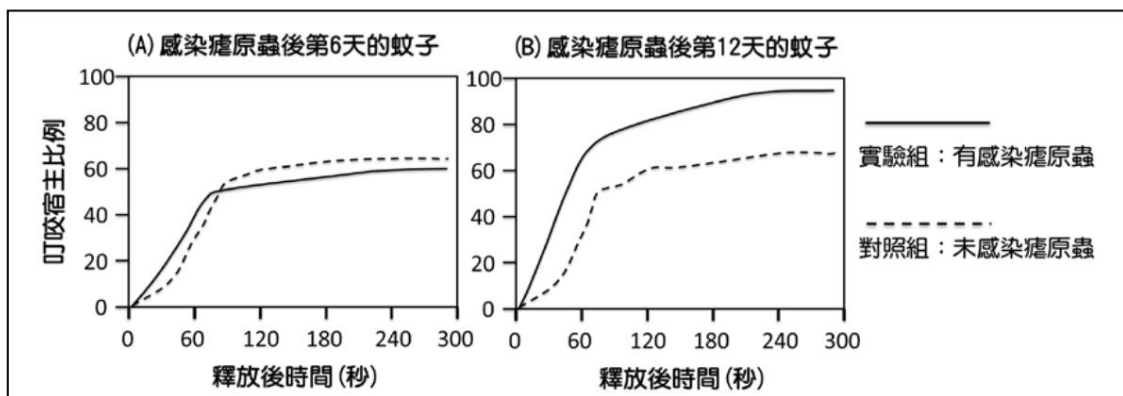
Which periods are most closely related to the occurrence of malaria parasites?

- Scientists fed mosquitoes blood containing malaria parasites and then observed the mosquitoes on the 6th day after infection.

Foraging behavior on day 12 and day 12 (mosquitoes were released into a netted cage, and their behavior was recorded at different times).

The cumulative percentage of mosquitoes biting a host was used to obtain the following chart. Please compare the malaria rates in charts A and B.

Each protozoan corresponds to a specific stage in its life cycle (A: _____ B: _____)

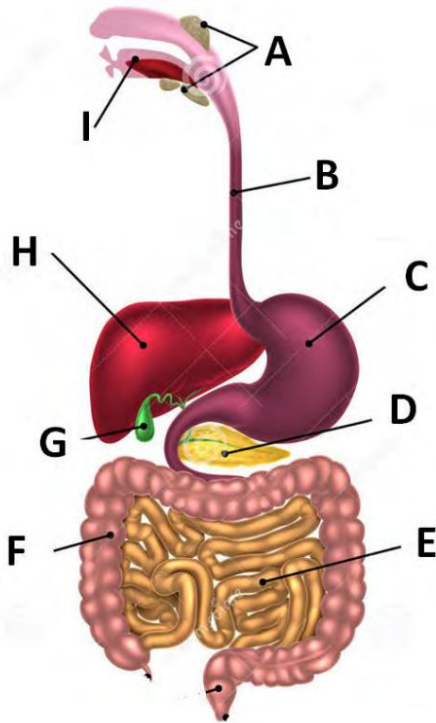


- (Continuing from the previous question) If the mosquito was infected with Plasmodium on the 6th day after infection, and the mosquito was infected on the 12th day after infection...

If the offspring exhibit the same foraging behavior, what impact will this have on malaria infection rates?

III. The lower left diagram is a schematic diagram of the human digestive system, and the lower right table represents enzymes or compounds. Use the diagram below in conjunction with...

Answer questions 1 and 2. (9 points)



code name	Enzymes or compound
	Amylase
	Ptyalin (salivary amylase)
	Trypsin
Man	Maltase
	Aminopeptidase
	Bile salts
	Heptanoic acid or gastric acid (HCl)
Spicy	Lipase
	Pepsin

Fill in the blanks in the following questions with English letter codes (representing organs) and Chinese character codes (representing enzymes or compounds).

The spaces in the text contain: (Not all code names will be used; some code names may be used more than **once**.)

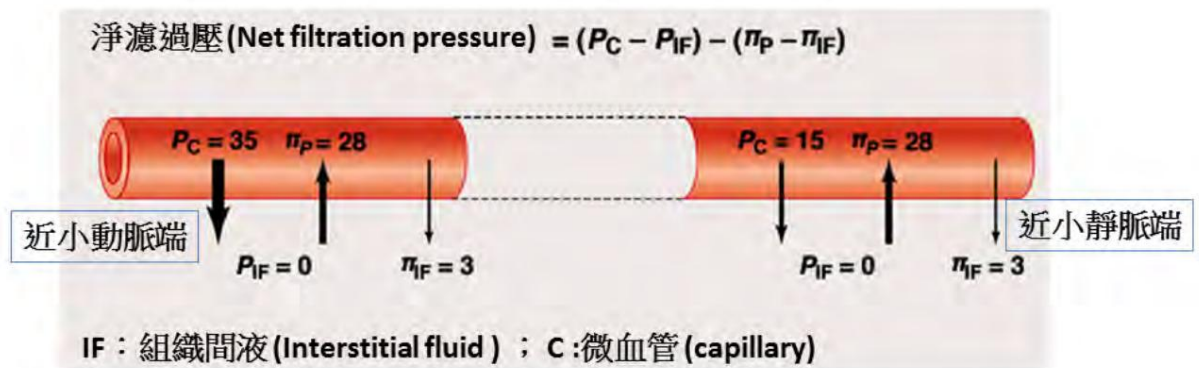
1. In organ 1a, fat is broken down by enzymes (or compounds) 1b and 1c into...
Fatty acids and glycerol, these enzymes or compounds are produced by...
Produced by official 1d. (4 points)

2. Proteins are first digested in organ 2a by enzymes (or compounds) 2b.
This enzyme or compound must first be activated by enzyme or compound 2c before it can be eliminated.
Proteins are broken down. The products of digestion are then further digested by enzymes (or compounds) into proteins.
Short-chain peptides (or oligopeptides), and these enzymes or compounds are
produced by organelle 2e. (5 points)

Please fill in the answers in the table below.

1a	1b	1c	1d	
2a	2b	2c	2D	2e

IV. The following diagram illustrates the human microvessels and net filtration pressure. Answer the following questions based on this diagram. (8 points)



1. Write down the names and directions of four forces that affect net filtration overpressure.

(1 point for each item, 4 points in total).

Question Number	Name of force	Direction of action (outflow or inflow into capillaries)
(1)		
(2)		
(3)		
(4)		

2. This diagram is also applicable to describing filtration in the glomerulus.

(1) In the glomerular filtration process of a normal, healthy human body, which force is missing? (1 point)

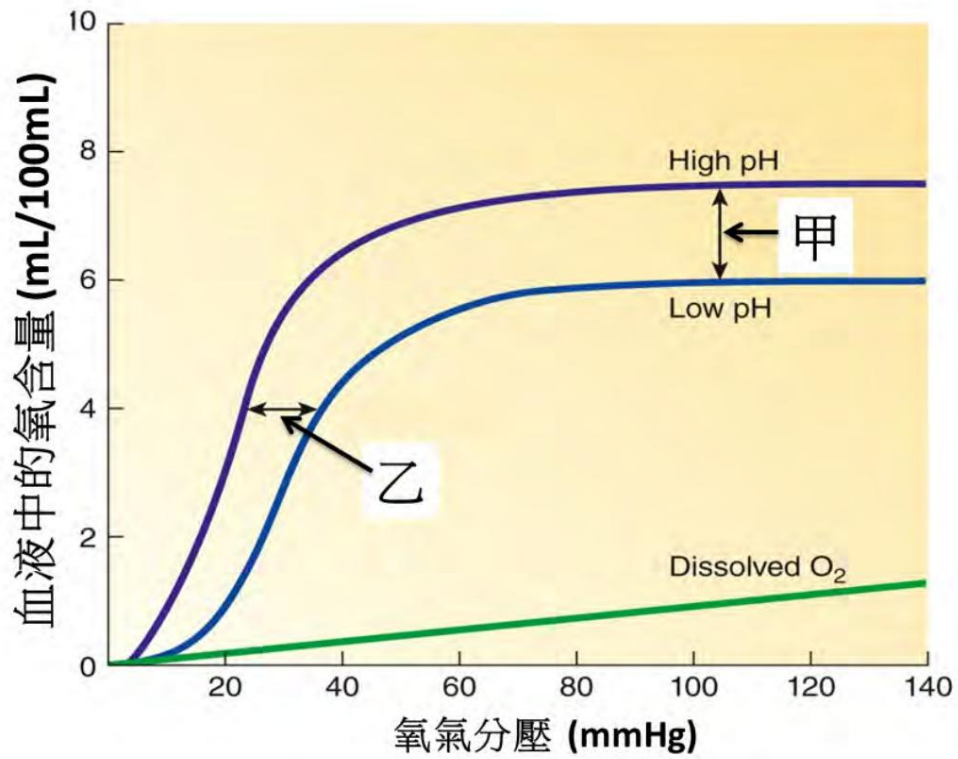
(2) Why is this force missing? (3 points)

V. The ability of human blood to carry oxygen varies in response to changes in physiological state, such as body temperature and pH levels.

The changes are as follows. The image below shows a normal, healthy individual responding to different oxygen partial pressures and pH levels.

Changes refer to variations in the oxygen content of the blood. "A" and "B" are two important physiological reactions.

Please answer questions 1 and 2 accordingly. (8 points)



1. Names of these two important physiological responses (1 point each, 2 points total)

2. The mechanisms of these two important physiological responses (3 points each, 6 points in total)

VI. Short Answer Questions (18 **points in total**)

1. Different nerve fibers can be classified based on the thickness of their axons and whether they are covered by a medullary sheath. Please compare.

These morphological differences relate to differences in neural function and physiological role playing. (6 points)

2. Nerve axons can be covered by Schwann cells or oligodendrocytes.

The myelin sheath forms many segments. If the myelin sheath were a complete, segmented structure, like the rubber coating on an electrical wire, it would be problematic.

What are the differences between a single axon and a structure that forms many segments? (6 points)

3. If nerves without myelin sheaths can also transmit action potentials, why do they only transmit them when injury causes nerve loss?

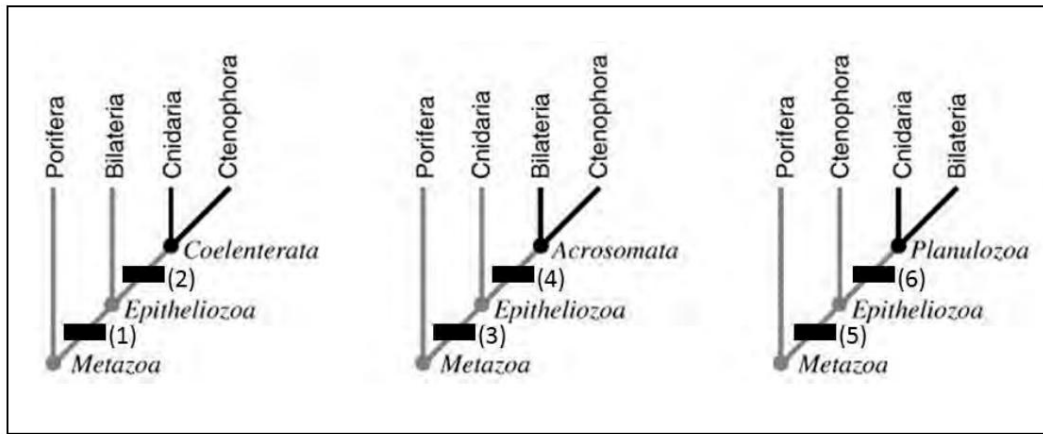
During myelination or in some acute inflammatory demyelination polyneuropathy processes, nerves lose their transmission capacity.

What is the function of generating electric potential? (6 points)

4. VX nerve agent poisoning often results in death due to respiratory muscle paralysis. Please explain its effect on nerve communication.

The mechanism of air transmission further explains the cause of respiratory muscle paralysis. (7 points)

VII. Questions and Answers



1. These are four taxa within the animal kingdom (Metazoa) (sponges, bilaterally symmetrical animals, etc.).

Three pairs of animals related to Bilateria, Cnidaria, and Ctenophora are mentioned.

I propose a hypothesis. What synapomorphic characters might support this hypothesis?

Is this true? Please fill in the possible co-derived features on these internodes below.

(3 points each , **18 points in total**)

(1) _____

(2) _____

(3) _____

(4) _____

(5) _____

(6) _____

2. Continuing from the previous question, assume that all three hypotheses are generated from the same matrix and by the parsimony method.

Given that the calculations and predictions are based on the same tree length, how would you decide which is the best hypothesis?

Say it? (**7 points**)

Theoretical Assessment Questions for the 2017 International

Biology Olympiad

National Team Selection **Camp**

Part Four

Student Code



Precautions:

1. This section of the test consists of 4 pages (including the cover) and is worth 100 points. Please check if you have it.
If any pages are missing, please inform the test administrator immediately.
2. When answering the questions, please check that the cover of the test paper matches your test number.

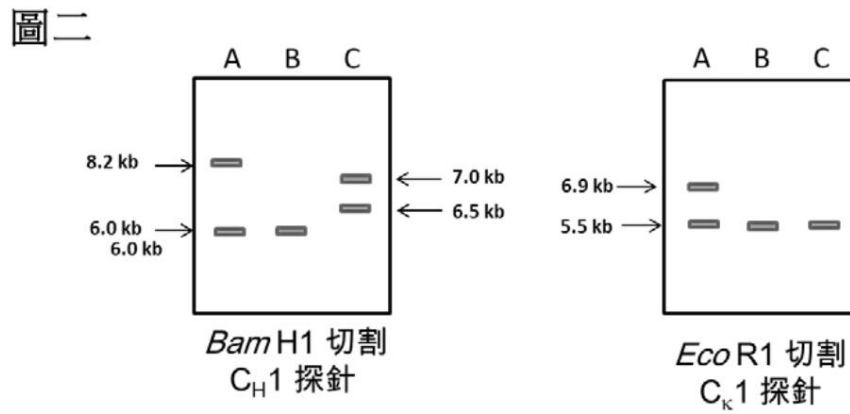
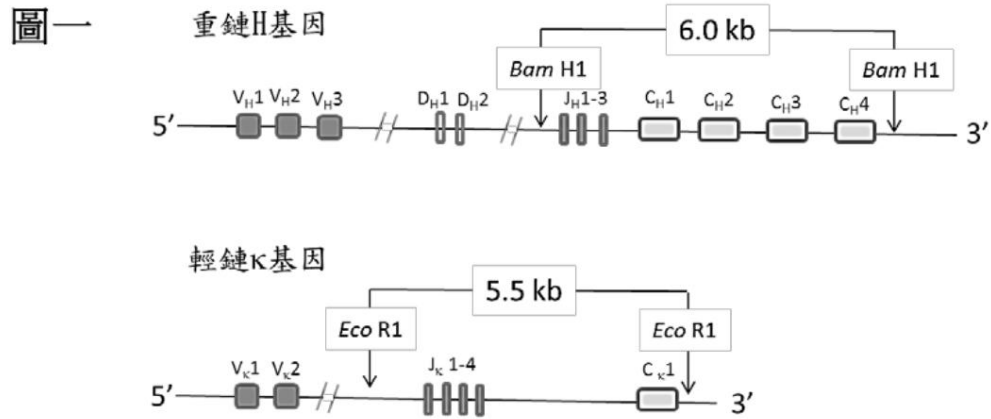
3. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.
4. You may write your answers on the back of the questions, but please indicate the question number.

Mammalian antibodies consist of four protein chains: two heavy chains (H) and two light chains (L). The binding site to the antigen is a highly variable region (V), while the remaining regions constitute the immutable region (C). The variable region of the heavy chain is determined by three gene segments called V, D, and J, while the light chain has only two genes: V and J. In humans, for example, there are approximately 20 different V genes, 20 different D genes, and 4 J genes. Furthermore, antibody gene expression exhibits allelic exclusion, meaning that only one of two paired genes is expressed. The 1987 Nobel Prize in Physiology or Medicine was

awarded to Dr. Susumu Tonegawa for his 1976 paper published in the Proceedings of the National Academy of Sciences, which explained the genetic theory that the immune system can produce diverse antibodies. His experiments using mice as a model indicated that during mammalian development, different genes combine randomly, resulting in thousands of distinct variable regions on the genetic material. Since each antibody has four variable regions, and genes themselves are inherited randomly, antibodies have the potential for hundreds of millions of variations. These results also confirmed the findings of W. Dreyer and C. Bennett of Caltech in the 1960s, who proposed that by randomly combining different proteins that constitute antibodies, millions of antibody variable regions could be generated to respond to potentially encountered antigens, successfully explaining the ability of the immune system to generate antibodies with limited...

The genes have the potential to create an infinite number of different

antibody molecules. When Xiao Ying, a first-year graduate student, first entered the lab, her advisor gave her three tubes of genetic DNA extracted from the liver, pre-B lymphoma cells, and IgM antibody-secreting plasma cells of the rare mammal Chinchilla. Because the tags had fallen off, the identities of these DNAs could not be identified, and Xiao Ying was instructed to design an experiment to distinguish these three types of DNA. Xiaoying carefully examined the gene structure map of the heavy and light chains of the Duoduo Chinchilla antibody, as well as the restriction enzyme cleavage sites and the numerical display of the cleaved DNA fragment sizes, shown in Figure 1. She also found CH1 and C γ 1 DNA probes labeled with isotopes in the laboratory freezer. Xiaoying labeled the three tubes of DNA as samples A, B, and C, and after completely cleaving them with *Bam* H1 and *Eco* R1 restriction enzymes, she performed agar gel DNA electrophoresis. The DNA in the gel was then transferred to cellulose acetate membranes, denatured, and hybridized using CH1 and C γ 1 DNA probes, respectively. Figure 2 shows the experimental results of Xiaoying's Southern Ink Dot Method analysis.



1. Please explain to Xiaoying what type of cells are in samples A, B, and C. (40 points)

2. Besides **DNA** recombination, what other molecular mechanisms of gene expression can lead to antibody molecule diversity?

(30 points)

3. What modifications and assembly processes are required after antibody protein production before it is secreted to form a functional antibody molecule?

(40 points)

Part Five

Student Code



4. You may write your answers on the back of the questions, but please indicate the question number.

Part Five: (100 points)

Multiple choice questions (3 points each , **24** points in total)

I. Grassland ecosystems cover a vast area of the earth's surface (approximately 40%), thus playing a crucial role in carbon accumulation and biodiversity.

It is extremely important for the maintenance and production of forage. The differences in the species composition of grasses can generally be divided into C3...

Grassland, C4 grassland and mixed grassland.

1.(For C3 and C4 grasslands, the net primary productivity per unit area is...

production , The order of NPP sizes is as follows:

(A) C4 grassland > C3 grassland

(B) C3 grassland > C4 grassland

2.(Two major trends resulting from global climate change are the increase in atmospheric carbon dioxide concentration.

The rising global temperatures and the resulting increase in carbon dioxide concentrations are impacting C4 and C3 grasslands .

The original impact is:

(A) The photosynthetic efficiency of C3 plants in grasslands increases accordingly.

(B) The photosynthetic efficiency of C4 plants in grasslands increases accordingly.

3.(The impacts of global climate change and rising temperatures on C4 and C3 grasslands are as follows:

(A) The photosynthetic efficiency of C3 plants in grasslands increases accordingly.

(B) The photosynthetic efficiency of C4 plants in grasslands increases accordingly.

4.(At the end of the Miocene (6-8 million years ago), grasslands rapidly expanded in tropical and temperate regions.

Zhang became an important global biota, which may be related to the high concentration of carbon dioxide in the atmosphere at that time.

The degree is relevant. The type of this grassland is:

(A) C4 grassland

(B) C3 Grassland

5.(Grasslands where C3 and C4 species coexist include those in South and North America, Central Asia, South Africa, and Australia.

In regions like New Zealand, the balance between C3 and C4 vegetation types in these grasslands will affect...

Primary productivity of grasslands, diet of herbivores, decomposition rate of roots and litter,

Nutrient cycling and other ecological functions. Under future global changes, their balance may be affected.

In addition to carbon dioxide concentration and temperature, the factors also include several other factors, among which the most important are...

The important ones are:

(A) Soil nitrogen content

(B) Soil moisture

(C) Fire

Second, tropical forests play a vital role in the global carbon cycle because they account for a significant portion of global primary production.

Half of the Gross Primary Production (GPP) contains 40% of the global carbon stock. But...

For tropical forests, several factors influence their carbon sequestration, including community composition, disturbance history, and Succession stage, climate, soil and other factors.

- 1.(In terms of the successional stage, the total primary production per unit area is:
 - (A) Secondary forest > Primary forest
 - (B) Primary forest > Secondary forest
- 2.(In terms of the successional stage, its net primary productivity per unit area is:
 - (A) Secondary forest > Primary forest
 - (B) Primary forest > Secondary forest
- 3.(Currently, net primary productivity per unit area of an ecosystem is measured.

The most common method for Production, NEP is:

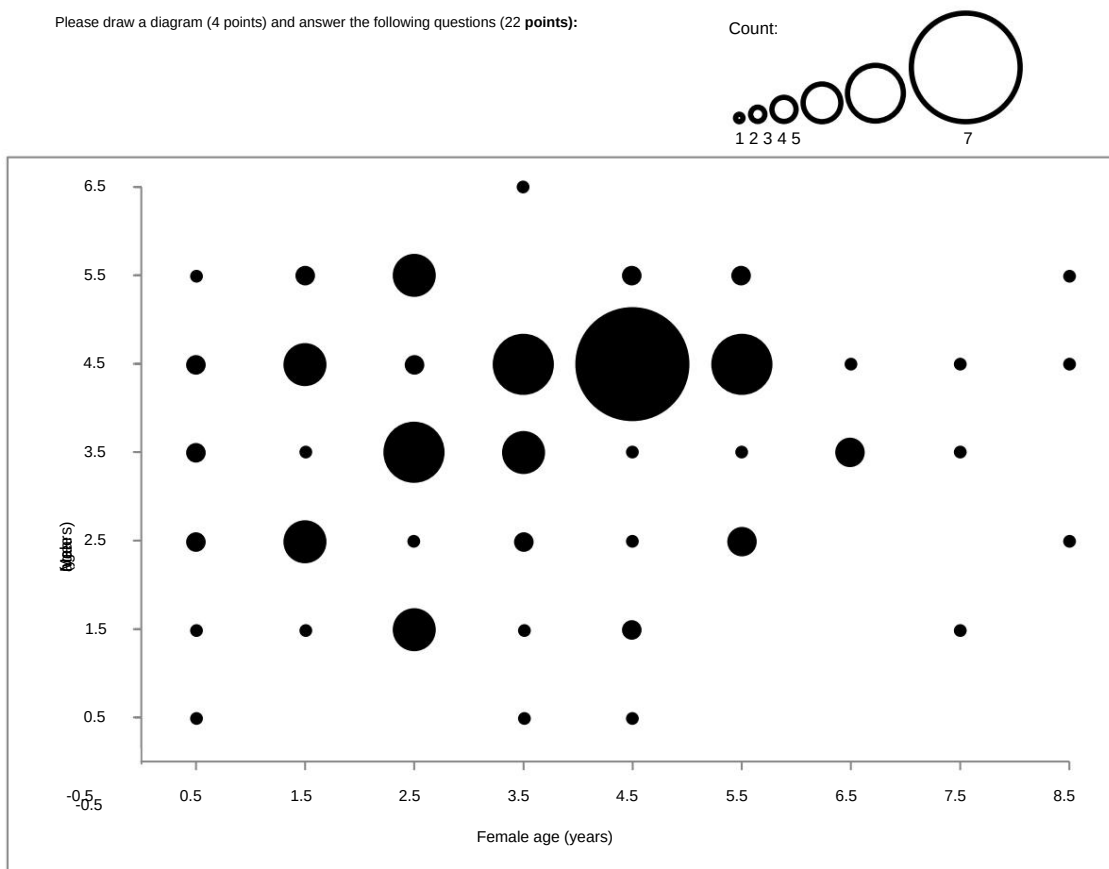
 - (A) Directly measure the annual biomass produced in the field (such as trunk diameter, litter weight)
(Quantity, etc.)
 - (B) Measure the flux of carbon dioxide entering and leaving the ecosystem using instruments (per unit area and per unit volume).
(the amount of data circulating in a given time)
 - (C) Measure the oxygen flux (per unit area and per unit hour) entering and leaving the ecosystem using instruments.
(Inter-channel flow)

Fill in the blanks and true/false questions (26 points in total)

III. The image below shows the observation and analysis of free-roaming white-tailed deer's reproduction over six years in a breeding park in the southern United States.

Based on the records of successful mating during breeding, try to create an age distribution chart (bar chart) for the deer based on the data shown in the figure.

Please draw a diagram (4 points) and answer the following questions (22 points):



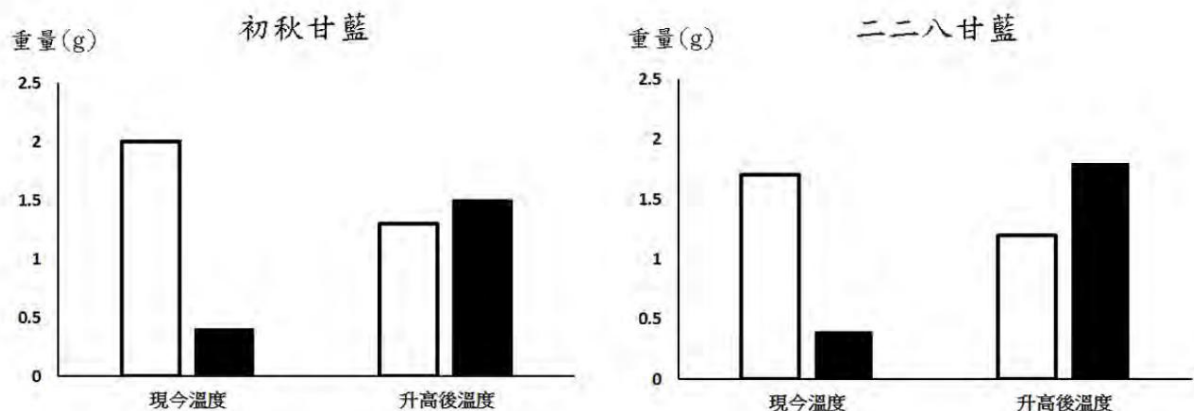
1. How many pairs are there in the picture? (2 **points**)
2. How many pairs of people of the same age are there? (2 **points**)
3. How many pairs have males older than females? _____ (3 **points**)
4. The largest age difference between the sexes is _____ years (1 point), and there are _____ pairs (1 **point**).
5. Among non-same-age pairs, the age difference that occurs most frequently is _____ years (1 point), and there are _____ pairs (1 **point**).
6. The average age of males is _____ years old (2 decimal places) (3 **points**)
7. The largest number of individuals in this herd across all age groups should be those under one year old. (1 **point**) (True/False)
8. The smaller the sex ratio in this herd, the lower the chance of polygamy. (1 **point**) (True/False) 9. The physical development of a young male pairing should be better than that of a female pairing of the same age. (1 **point**) (True/False) 10. Sexual selection can be observed in this population. (1 **point**) (True/False) 11. The reason why older males mate with younger females may be that they produce more offspring. (1 **point**) (True/False) 12. The mortality rate of males is higher than that of females in this population. (1 point) (True/False) 13. This study mainly relies on long-term observation of the behavior of deer in the wild during the breeding season. (1 **point**) (True or False)
14. The length of the breeding season may affect the pairing outcome (1 **point**) (True/False)

True or False and Drawing Questions (25 points in total)

IV. Cabbage (Chinese cabbage) prefers to grow in cooler climates. While its cultivation is suitable for Taiwan's plains during the lower winter temperatures, the large planting area in the plains often leads to low prices during peak winter harvests. Conversely, in summer, when temperatures in Taiwan's plains are too high, resulting in poor quality (taste) of cabbage, the mountainous areas of Taiwan, with their lower temperatures, remain suitable for cabbage cultivation. However, the smaller planting area and lower yield in the mountains often lead to higher prices for higher-quality cabbage in the summer. This incentivizes farmers to continue cultivating cabbage in the mountains, or even increase the planting area, creating a conflict between soil and water conservation and economic development in the mountains. To estimate whether the continued rise

in global temperature and carbon dioxide concentration will affect the yield of Taiwan's high-altitude cabbage, making it increasingly unsuitable for cabbage cultivation in Taiwan's mid-to-high-altitude mountains, researchers planted cabbage seedlings in four growth boxes with four different combinations of atmospheric temperature and carbon dioxide concentration, while keeping other environmental conditions, such as light cycle and humidity, consistent. These four growth chambers provided the following four environments: 1. Current atmospheric temperature and current atmospheric carbon dioxide concentration; 2. Current atmospheric temperature and increased atmospheric carbon dioxide concentration; 3. Increased atmospheric temperature and current atmospheric carbon dioxide concentration; 4. Increased atmospheric temperature and increased atmospheric carbon dioxide concentration. Regarding increased temperature and increased carbon dioxide concentration, according to the 2013 report by the Intergovernmental Panel on Climate Change (IPCC), it is estimated that global temperatures could increase by 4°C by 2100, with carbon dioxide concentration increasing to 936 ppm (currently around 400 ppm). For the cabbage varieties, researchers selected "Early Autumn" cabbage, commonly grown in mid-to-high altitude mountainous areas, and "228" cabbage, commonly grown in lowland areas in summer, which is more heat-resistant but has a poorer taste, to compare whether different cabbage varieties respond differently to increases in temperature and carbon dioxide concentration. Researchers placed 10 early autumn cabbage seedlings and 10 228 cabbage seedlings of similar height in each growth box and allowed them to grow freely. After two months, the seedlings were removed, and the weight of each seedling was recorded. The average weight of the 10 seedlings of the same cabbage variety in each growth box was then calculated. The final average values are shown in Figure A (the standard deviation of each average value is not shown for simplicity).

(A)



The hollow column represents the current atmospheric carbon dioxide concentration, while the solid column represents the carbon dioxide concentration after the increase.

Based on the results in Figure A , answer the following questions (true or false, **3** points each):

- (**1.** When the atmospheric temperature rises, the weight of cabbage seedlings will decrease in early autumn.
- (**2.** When the atmospheric temperature rises, the weight of 228 cabbage seedlings will decrease.
- (**3.** When the concentration of atmospheric carbon dioxide increases, the weight of the 228 cabbage seedlings also increases.
- (**4.** When both atmospheric temperature and atmospheric carbon dioxide concentration increase, the weight of cabbage seedlings will decrease in early autumn.
- (**5.** When both atmospheric temperature and atmospheric carbon dioxide concentration increase, the weight of 228 cabbage seedlings will decrease.

Suppose researchers placed different varieties of cabbage seedlings into four identical growth boxes, and after two months...

The following average weight (g) was then obtained:

	Current atmospheric temperature	Increased atmospheric temperature
The current atmospheric carbon dioxide concentration	3	1.5
has increased.	2	0.5

Based on the information in the table above, please draw a similar diagram B, modeled after diagram A. (**6 points**)

Based on your drawing B, answer the following questions (true or false, **2** points each):

- (**1.** When the atmospheric temperature rises, the weight of this variety of cabbage seedlings will decrease.
- (**2.** When the atmospheric carbon dioxide concentration increases, the weight of this variety of cabbage seedlings will decrease.

True or False (10 questions, 25 points in total)

5. The pepper moth (*Biston betularia*) is widely distributed in the temperate zones of the Northern Hemisphere.

A type of moth, also quite numerous in Britain. From 1850 onwards...

Previously, pepper moths in Britain were almost all light-colored, with body colors primarily consisting of...

Gray with irregular dark brown spots. Discovered in 1848.

Now, the dark-colored pepper moth has appeared in some parts of Britain by 1895.

The number of dark-colored pepper moths in the area has increased dramatically, even accounting for a significant portion of all pepper plants.

98% of the moths are of this type. Additionally, some pepper moths possess an intermediate form.

body color.

JW Tutt believes that the trunks and branches of British trees were originally covered with moss and lichen, which explains the lighter-colored pepper.

Moths find excellent camouflage when perched on it, making them difficult for birds to spot. The Industrial Revolution led to Britain's rise in the 19th century...

The burning of large amounts of coal causes sulfur dioxide and soot in the air to adhere to the tree trunks, leading to the death and discoloration of mosses and lichens.

The color changes to black, giving the dark-colored pepper moths good camouflage, which allows them to dominate in number.

In the 1950s, H.B. Kettlewell conducted a series of experiments in Birmingham, England.

In one experiment, he chose a polluted forest near an industrial area and used a tag-release-recapture method.

This method was used to study the population variations of pepper moths of different color types. Kettlewell used pepper moths captured in the wild and those cultured in the laboratory.

The moths were marked on the abdomens of different colored pepper moth morphs with a harmless dye, and the marked moths were released each morning.

The 630 birds were placed on tree trunks and allowed to roam freely until they settled quietly on the trunks. Then, that same night...

Mercury lamps and pheromone traps were used to capture pepper moths (770 in total) in the forest. The experiment lasted for a total of 11 days.

On that day, the total number of pepper moths captured and the number of tagged pepper moths were calculated, and the results are listed in the table below.

time (1953)	Release (marked)			total number	Total number of moths captured			total number	Capture the tagged			total number	
	moth numbers				moth numbers				moth numbers				
	deep	shallow	middle		deep	shallow	middle		deep	shallow	middle		
	color	color	between		color	color	between		color	color	between		
	type	type	type		type	type	type		type	type	type		
June 25th 10		12	10	32	8	0	1	9	-	-	-	-	
June 26th 0		0	0	0	127	15	7	149	3	1	1	5	
June 27th 33		11	15	59	34	5	1	40	1	0	1	2	
June 28th 37		21	5	63	23	3	3	29	2	0	2	4	
June 29th 0		0	0	0	55	10	1	66	5	4	0	9	
June 30th 68		26	8	102	37	3	2	42	1	0	1	2	
July 1st 90		21	3	114	76	9	2	87	19	2	2	23	
July 2nd, 74		21	3	98	75	13	4	92		6	0	34	
July 3rd, 68		15	0	83	77	11	10	98	25	3	1	29	
July 4th 67		10	2	79	66	9	2	77	23	2	0	25	
July 5th 0		0	0	0	73	3	5	81	16	0	0	16	
total	447	137	46	630	651		81	38	770	123	18	8	149

Source: Kettlewell, HBD 1955. Heredity (9):323-342.



Please answer the following questions based on the statements below. (True or False, **2.5** points each)

((A) This experiment can verify that the color of the pepper moth is controlled by genetic genes.

((B) Genetic drift can cause the color ratio of pepper moths to completely reverse within 40 years.

((C) This experiment supports the high survival rate of black pepper moths in polluted forests.

((D) This experiment supports the view that the intermediate pepper moth has a lower survival rate than the black pepper moth in polluted forests.

Pepper moth.

((E) This experiment can verify that birds are the main predators of the pepper moth in the UK.

((F) The movement of individuals of pepper moths that originally lived in the study site may affect the experimental results.

((G) If the same experiment is conducted in an unpolluted forest (with light-colored tree trunks), the light-colored beech can be expected to...

The survival rate of the pepper moth is relatively high.

((H) Excessive black particles in the environment may cause pepper moths to absorb more black particles, thus affecting their body size.

The color darkened, and this experiment cannot rule out this possibility.

((I) If environmental pollution is brought under control, the tree trunk color will turn lighter, and the number of light-colored pepper moths will be higher than that of other species.

The example should improve.

((J) If environmental pollution is brought under control, the tree trunk color will turn light, pale, intermediate, or dark.

The ratio of pepper moths should be close to 1:2:1.